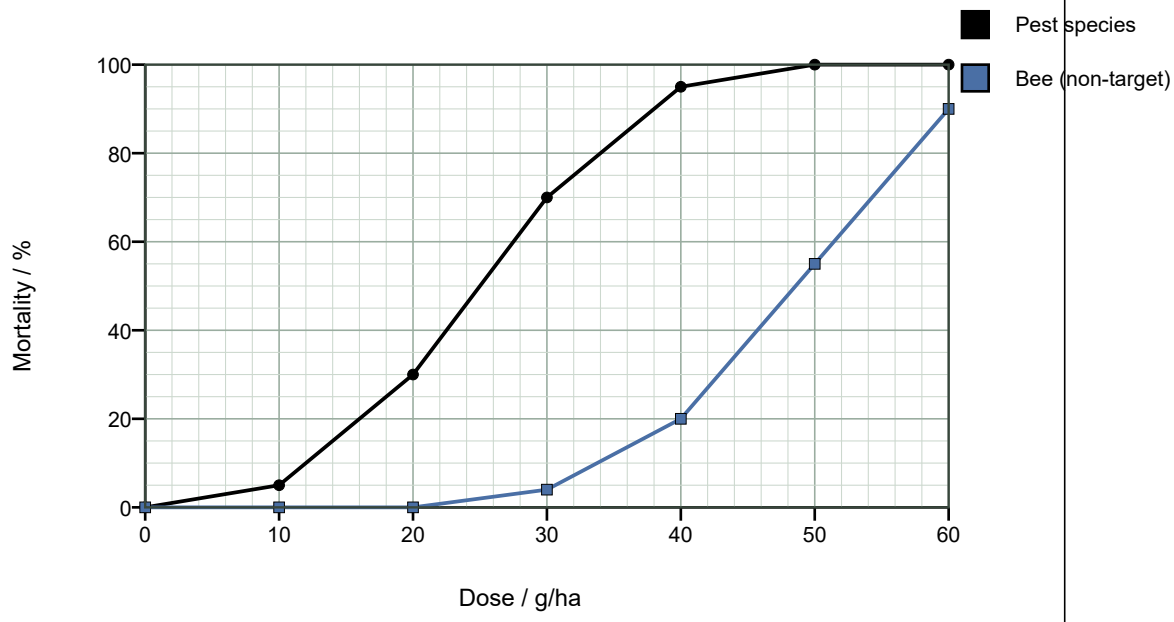
Questions by topic: Pesticides - Set 1 (70 marks)

Answer all questions in the spaces provided.

0 1 Figure 1 shows how the percentage mortality of a pest species and of a non-target bee species change with the dose of an insecticide applied to a crop.

Figure 1



0 1 . 1 Use Figure 1 to estimate the LD_{50} of the insecticide for the pest species.

[1 mark]

Answer ____ g/ha

0 1 . 2 Define the term **MDNF** (maximum dose never fatal).

[1 mark]

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0 1 . 3 Use Figure 1 to explain why it may not be possible to kill all of the pest species without harming bees.

[3 marks]

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0 1 . 4 Suggest how the farmer could reduce the harm to bees without reducing the dose applied.

[2 marks]

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0 1 . 5 Explain why a pesticide with high specificity is less likely to cause environmental damage.

[3 marks]

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0 2 Table 1 shows some properties of four groups of insecticide.

Table 1

Insecticide group	Toxicity to insects	Toxicity to mammals	Persistence	Liposolubility
Organochlorines (eg DDT)	High	Low	High	High
Organophosphates (eg malathion)	High	High	Low	Low
Pyrethroids	High	Low	Low	Low
Neonicotinoids	High	Low	Moderate	Low (water soluble)

0 2 . 1 Use Table 1 to explain why organochlorine insecticides caused the decline of birds of prey such as the peregrine falcon in the 1950s and 1960s.

[4 marks]

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0 2 . 2 Organophosphates replaced organochlorines for many uses. Use Table 1 to give **one** advantage and **one** disadvantage of this change.

[2 marks]

Advantage .....

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Disadvantage .....

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0 2 . 3 Explain why pyrethroids should not be used near rivers or fish farms.

[1 mark]

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0 2 . 4 Describe how the properties of neonicotinoids allow them to be used as a seed coating, and explain why this method of application reduces harm to non-target species.

[3 marks]

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0 3 A pesticide was applied to a field of oilseed rape. Table 1 shows the concentration of the pesticide in organisms sampled from a pond next to the field.

Table 1

Organism	Trophic level	Pesticide concentration / mg kg^{-1}
Pond water	-	0.002
Phytoplankton	1	0.04
Water fleas (Daphnia)	2	0.5
Small fish	3	4.2
Heron	4	38.0

0 3 . 1 Calculate how many times more concentrated the pesticide is in the heron than in the pond water. Give your answer to two significant figures.

[2 marks]

Answer ☐ times

0 3 . 2 Name the process shown by the data in Table 1.

[1 mark]

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0 3 . 3 Explain the **two** properties of the pesticide that cause the pattern shown in Table 1.

[4 marks]

0 3 . 4 The pesticide was probably an organochlorine. Suggest why the use of organochlorines is now banned or heavily restricted in most countries, even though DDT is still used to control malaria mosquitoes in some houses.

[3 marks]

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0 4 . 1 Neonicotinoids are the most widely used group of insecticides. Explain how the properties of neonicotinoids affect their impact on bee populations and describe the methods used to reduce this impact.

[9 marks]

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Extra space

0 4 . 2 State **one** way in which pesticide use in agriculture may be reduced by using natural processes.

[1 mark]

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0 5 Two insecticides, X and Y, were tested on a pest insect and on a predatory ground beetle that feeds on the pest. Table 1 shows the results.

Table 1

Insecticide	LD_{50} for pest / mg kg^{-1}	LD_{50} for ground beetle / mg kg^{-1}	Half-life in soil / days	Solubility in water
X	5	6	180	Low
Y	8	150	4	High

0 5 . 1 Explain what is meant by LD_{50}.

[1 mark]

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0 5 . 2 Using Table 1, explain which insecticide is more likely to cause indirect effects on the pest population through the food chain.

[3 marks]

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0 5 . 3 Suggest **two** environmental problems that could be caused by insecticide Y that are less likely with insecticide X.

[2 marks]

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2 .....

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0 5 . 4 Explain why the half-life of insecticide X makes it more likely to bioaccumulate.

[2 marks]

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0 5 . 5 Give **two** reasons why a systemic insecticide may be preferred to a contact insecticide.

[2 marks]

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2 .....

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0 6 . 1 Define the term **persistence** as applied to a pollutant.

[1 mark]

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0 6 . 2 Explain how the persistence of a pesticide is affected by **two** named environmental factors.

[4 marks]

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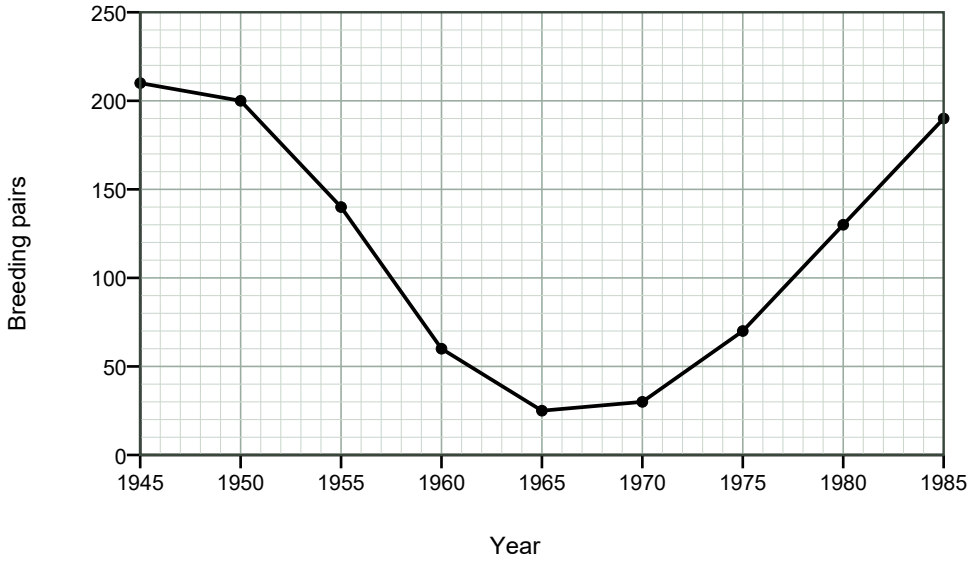
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0 7 Figure 1 shows the number of breeding pairs of sparrowhawks in a study area in the UK between 1945 and 1985, and the years in which DDT use began and was banned.

Figure 1



0 7 . 1 Describe the trend shown in Figure 1.

[2 marks]

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0 7 . 2 Calculate the percentage decrease in breeding pairs between 1950 and 1965. Show your working.

[2 marks]

Answer ..... %

0 7 . 3 DDT use in the UK began in the late 1940s and was restricted from the mid 1960s. Explain how the properties of DDT could account for the pattern in Figure 1.

[4 marks]

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0 7 . 4 Suggest why the recovery after 1970 was slow.

[2 marks]

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0 8 . 1 Which of the following properties would make a pesticide **least** likely to biomagnify? Tick one box.

[1 mark]

A High liposolubility and high persistence ☐

B Low liposolubility and low persistence ☐

C High liposolubility and low persistence ☐

D Low liposolubility and high persistence ☐

0 8 . 2 Explain the difference between **bioaccumulation** and **biomagnification**.

[2 marks]

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0 8 . 3 Explain why pesticides are described as an unusual type of pollutant.

[2 marks]

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END OF QUESTIONS

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